PROSUN BANERJEE

Software / AI Engineer | LLM Applications, Backend Systems, Agentic Workflows
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SUMMARY

Final-year Data Science student with three internships in backend, automation and full-stack engineering. Built and deployed an LLM-driven LinkedIn automation system (FastAPI, PostgreSQL, Playwright). Contributed to a multi-brand production CRM (Laravel) with role-based access control and tenant-isolation testing. Author of Rei, a local-first Windows voice assistant where model output is a proposal gated by a policy engine. Seeking SDE, AI Engineer and Agentic Systems roles.

EXPERIENCE

Full Stack Developer Intern, SkillEnable Edutech (Academy of Vedic Vidya) | Kolkata | Aug 2026 – Oct 2026

- Developed features for a multi-brand production CRM (Laravel) across 2 brands and 6 role tiers, shipping weekly releases against feedback from sales, marketing and academic teams.
- Built role-based access control: 4 independent CRUD permissions for Custom Reports, per-report Lead Insight access, and hidden/masked lead-field rules enforced across reports, dashboards, notifications, tasks, telephony and mobile.
- Tested tenant isolation across data, configuration and automation layers using 5 seeded leads, custom stages/fields and a live automation; observed no cross-brand leakage in these tests.
- Implemented a reminder and notification engine (1/3/5-day presets, custom and recurring reminders, logged email templates, scheduled daily digest replacing per-lead emails) and per-user call analytics (inbound, outbound, connected, unique leads, talk-time bands).
- Integrated Sentry and Interakt WhatsApp OTP; fixed Render memory exhaustion by capping Next.js build workers to 4; compared 100ms, Amazon Chime SDK and LiveKit on a 1,000 × 30-min call cost model.

Automation Engineer Intern, Teliolabs Communications (AI Konekts) | Remote | Apr 2026 – Aug 2026

- Designed and deployed a 6-module browser-automation system (Python, Playwright) on an Azure VM to run outreach across ~2,400 Apollo-sourced leads.
- Built an LLM engagement bot on Groq (Llama 3.3 70B) that generates context-aware comment replies, with rate limiting for sustained operation.
- Built a FastAPI backend with PostgreSQL and an analytics dashboard to track outreach and engagement performance.
- Also ran AI Konekts LinkedIn content (daily posts, two weekly newsletters).

Full Stack Development Intern, Prasarnet | Kolkata | Jun 2025 – Aug 2025

- Built an Exam Portal end to end on the MERN stack in an agile team.

PROJECTS

Rei: Local-First AI Operating Layer for Windows | Python, Pydantic, PySide6/QML, faster-whisper, Kokoro-ONNX, NVIDIA NIM, GGUF

- Designed around one rule: model output is a proposed intent, not a direct action. An 8-stage pipeline (capture -> propose -> parse -> decide -> confirm -> grant -> execute -> report) validates intents against Pydantic schemas. A policy engine returns ALLOW/CONFIRM/DENY, and execution requires a single-use grant token tied to a typed capability.
- Hybrid inference: cloud NVIDIA NIM (Llama 3.2 11B) under a 5-second deadline, with automatic fallback to a local Llama 3.2 1B (GGUF) so core functions work offline. Voice runs on-device: GPU faster-whisper STT, WebRTC VAD, Kokoro-ONNX TTS.
- Stored long-term memory in SQLite encrypted at rest with Windows DPAPI; enforced module layering with import-linter in CI; built a PySide6/QML control center and floating voice orb.

Plant Disease Classifier | PyTorch, CNN

- CNN that classifies plant disease from leaf images; built for Smart India Hackathon 2024 and presented at IEMHacks 2025.

SKILLS

Languages: Python, PHP, JavaScript
Backend & Data: FastAPI, Laravel, Node.js/Express, PostgreSQL, MongoDB, Pydantic
AI/ML: LLM integration (Groq, NVIDIA NIM), local inference (GGUF), PyTorch, faster-whisper, Kokoro-ONNX
Frontend: React, Next.js, PySide6/QML
Infra & Tools: Azure VM, Render, Sentry, Playwright, Interakt WhatsApp API, Claude Code
Concepts: RBAC, multi-tenant access control, API integrations

EDUCATION

BSc Data Science, Gurunanak Institute of Technology (MAKAUT) | 2023 – 2027 (expected) | CGPA 7.65 (6 semesters)

ACHIEVEMENTS

Finalist, National Entrepreneurship Challenge 2025 (national finals at IIT Bombay, representing college E-Cell)